

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
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S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 952+100

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

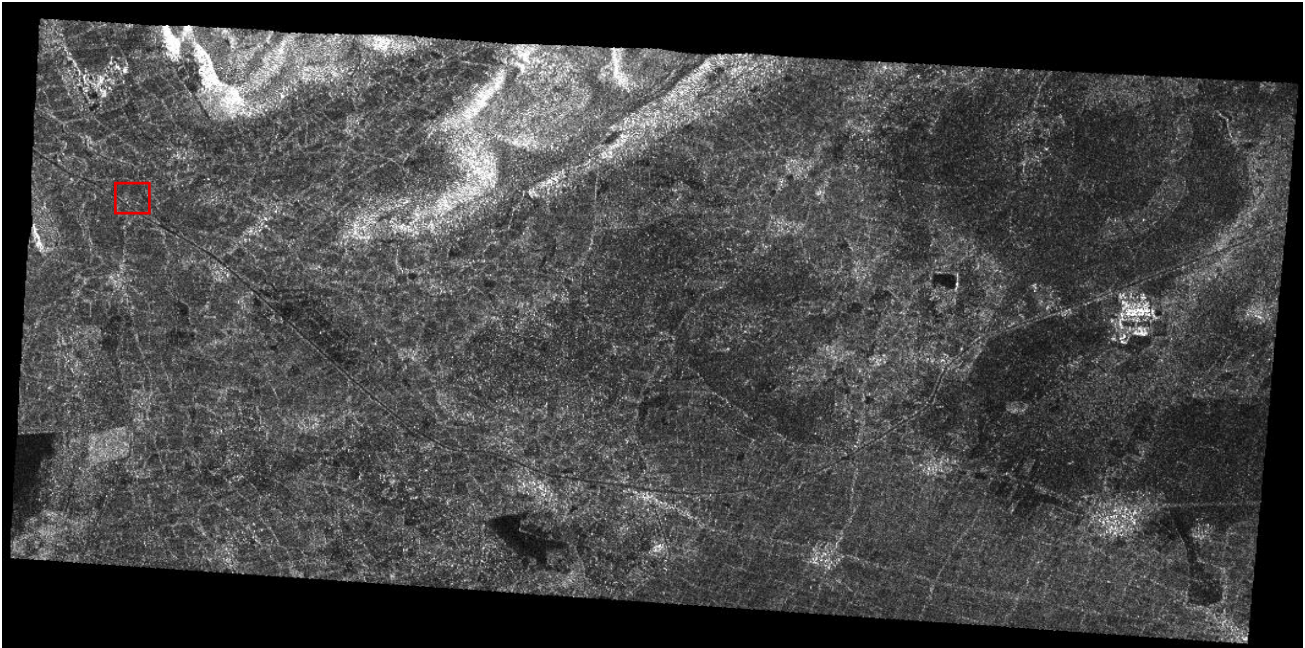
Flight Direction - DESCENDING

Polarization – VV+VH

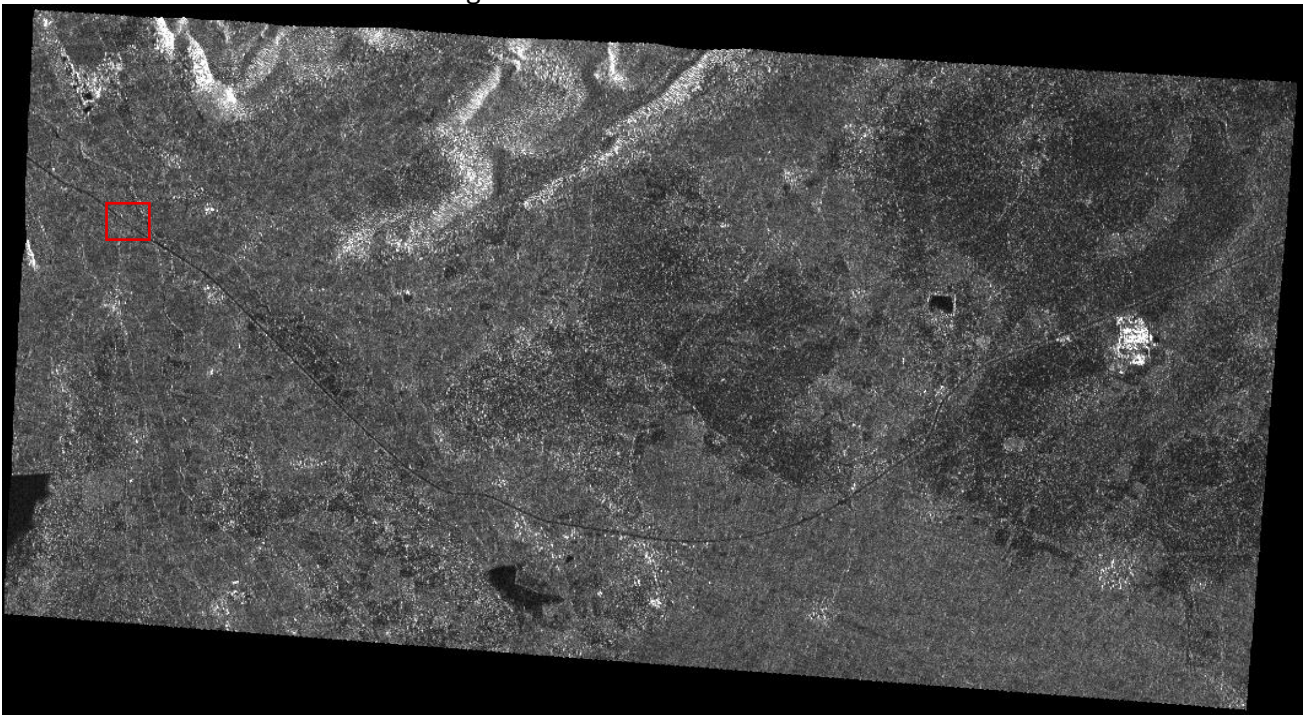
Absolute Orbit – 54706

2. Calibration, georeferencing, ortho rectification of SAR images (VH/VV)

- Ortho rectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

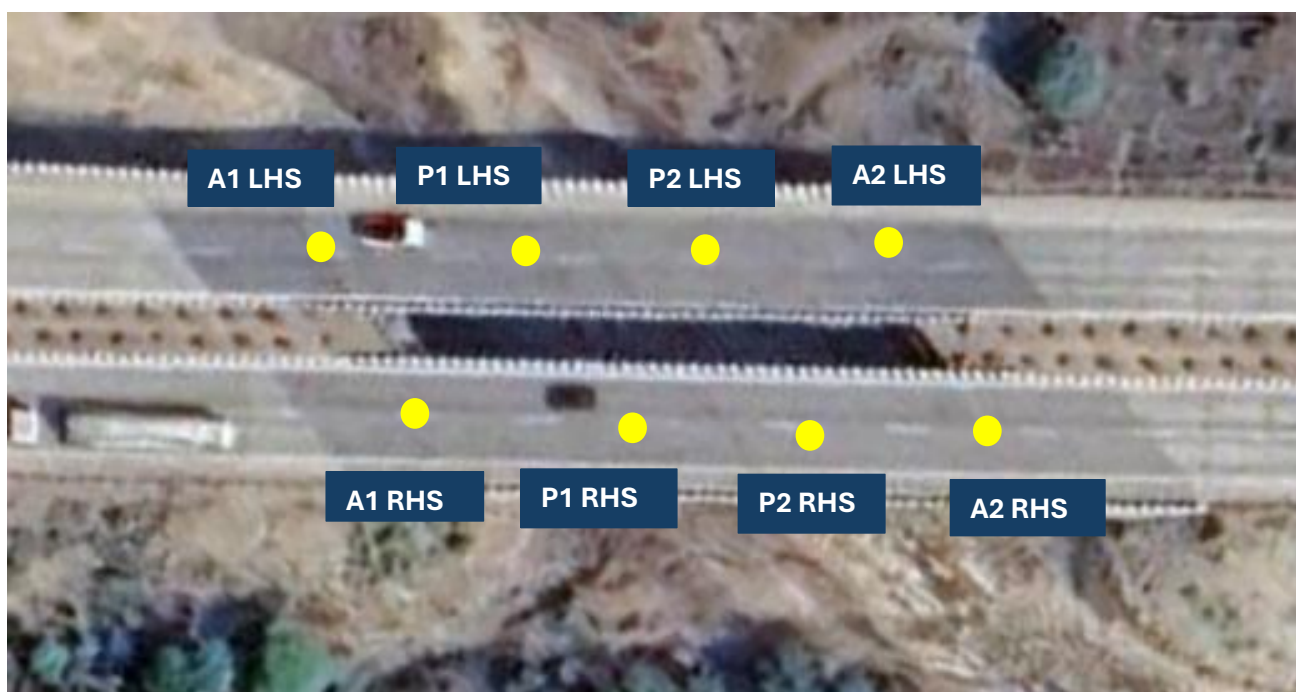


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-952+100

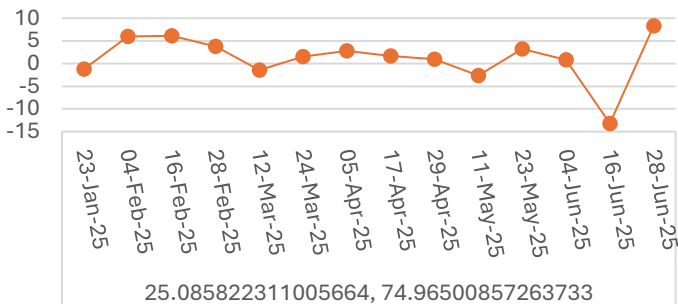
Length – 62.74M



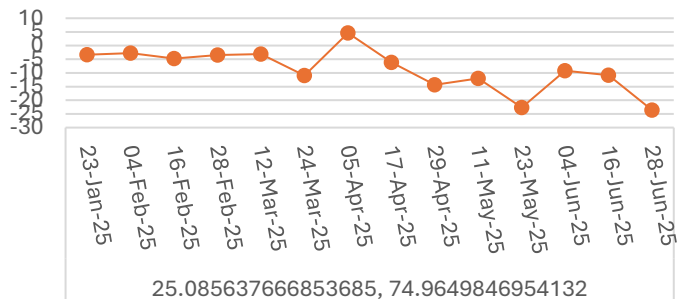
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
P2	Pier 2		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025

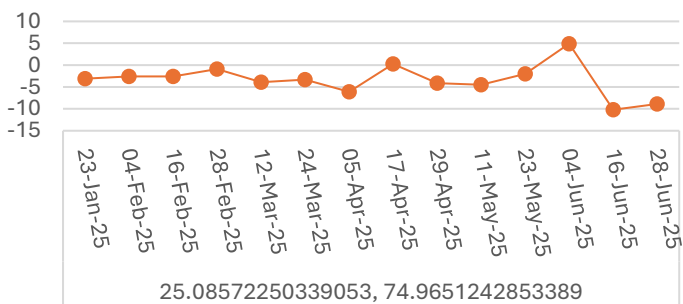
Displacement @ A1 LHS VV



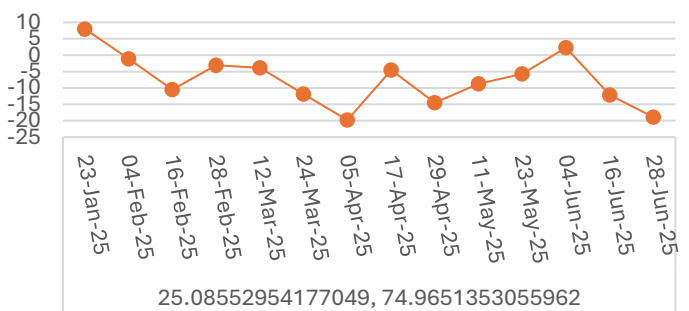
Displacement @ A1 RHS VV



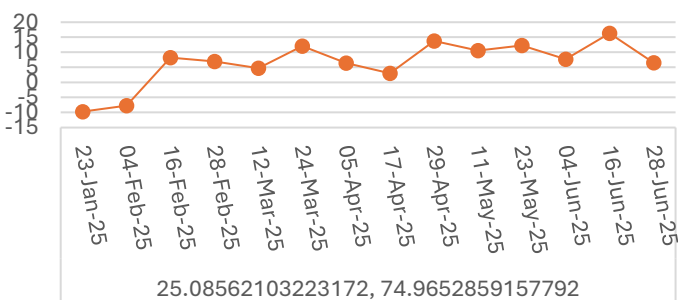
Displacement @ P1 LHS VV



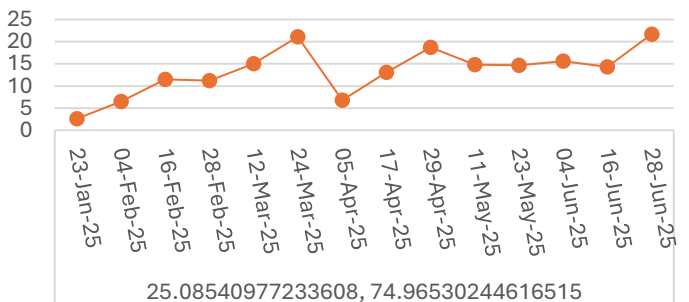
Displacement @ P1 RHS VV



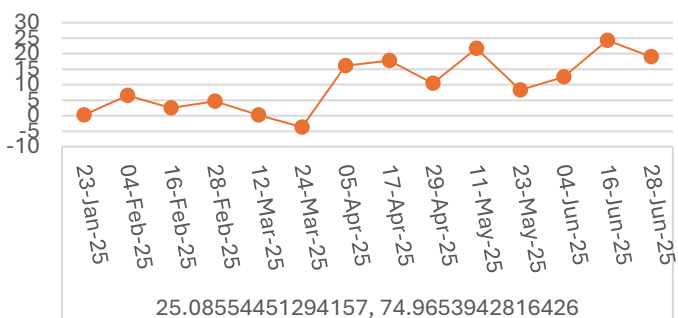
Displacement @ P2 LHS VV



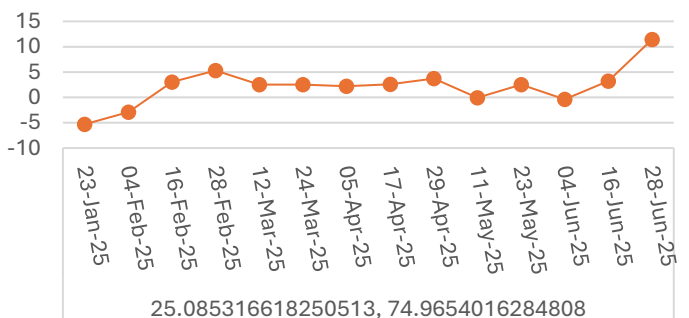
Displacement @ P2 RHS VV



Displacement @ A2 LHS VV

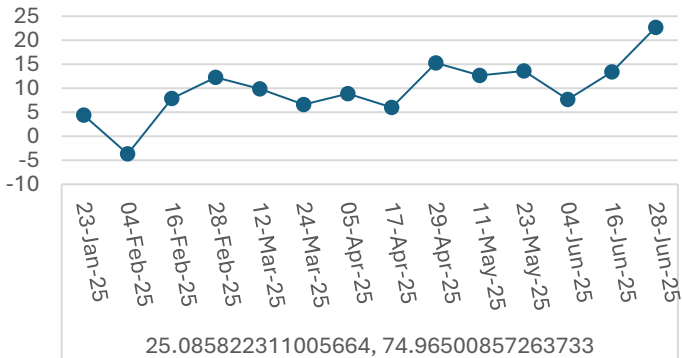


Displacement @ A2 RHS VV

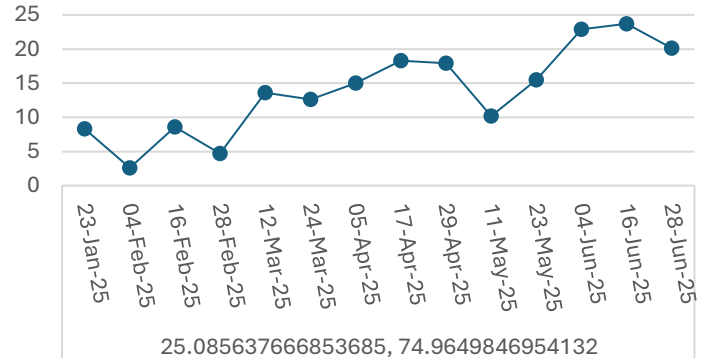


3.2 VH Displacement From 23 Jan 2025 To 28 June 2025

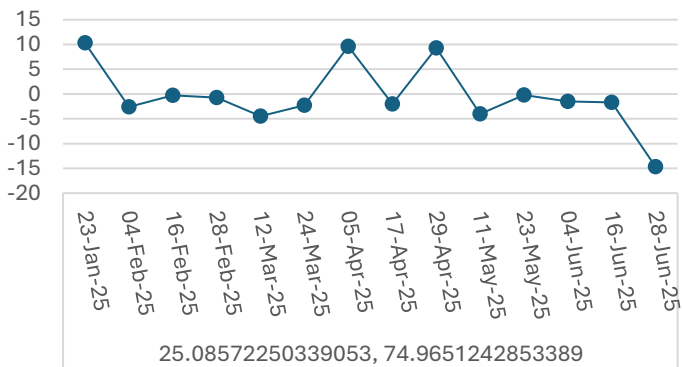
Displacement @ A1 LHS VH



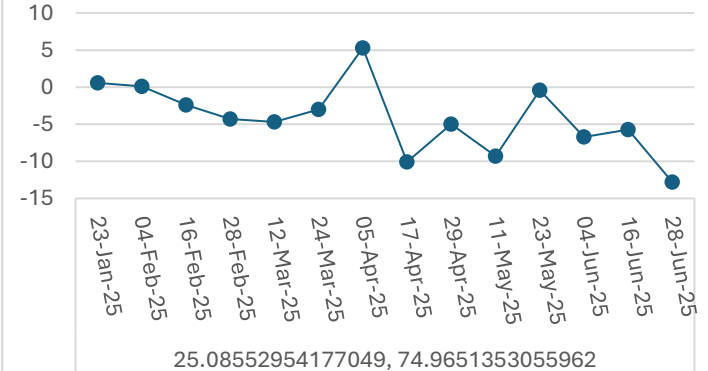
Displacement @ A1 RHS VH



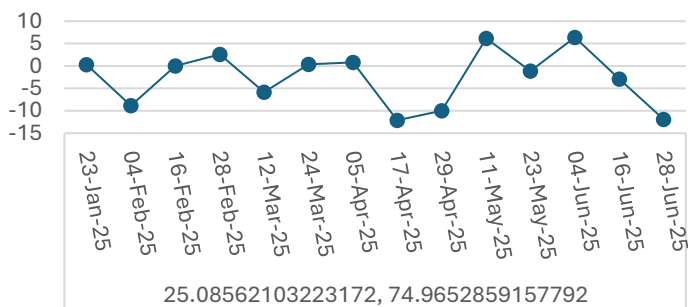
Displacement @ P1 LHS VH



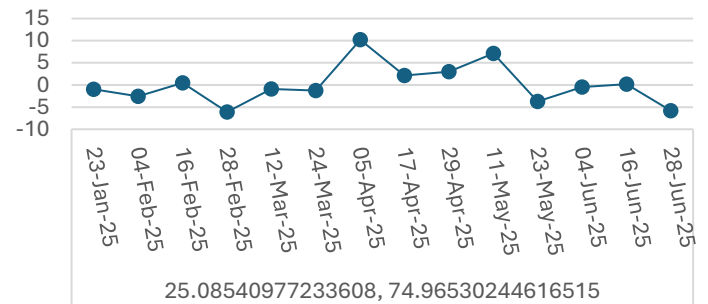
Displacement @ P1 RHS VH



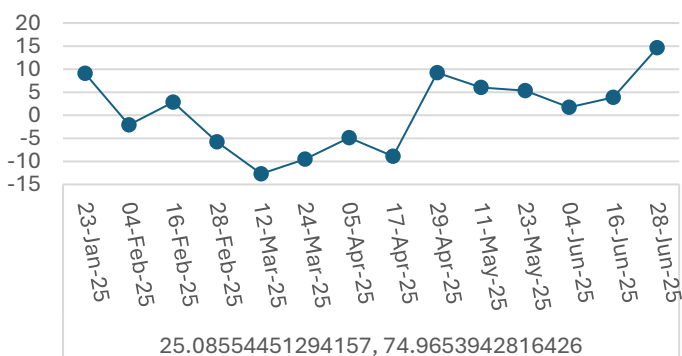
Displacement @ P2 LHS VH



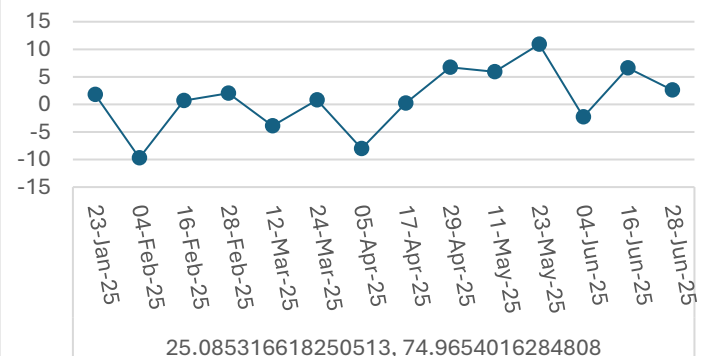
Displacement @ P2 RHS VH



Displacement @ A2 LHS VH



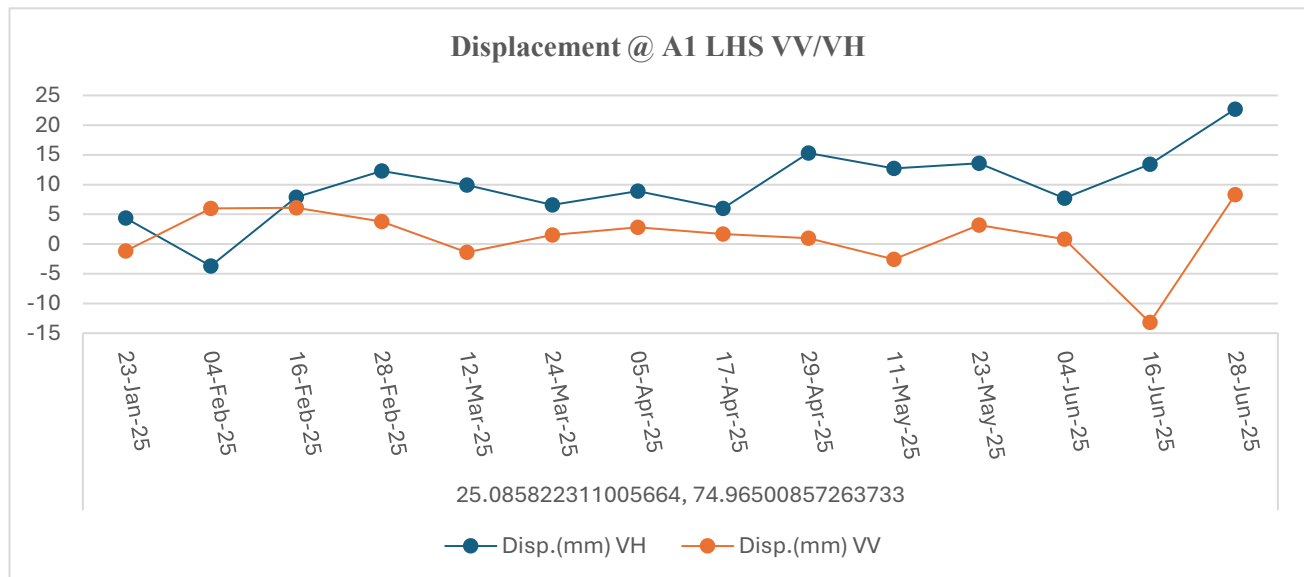
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.085822311005664, 74.96500857263733)



Key Observations:

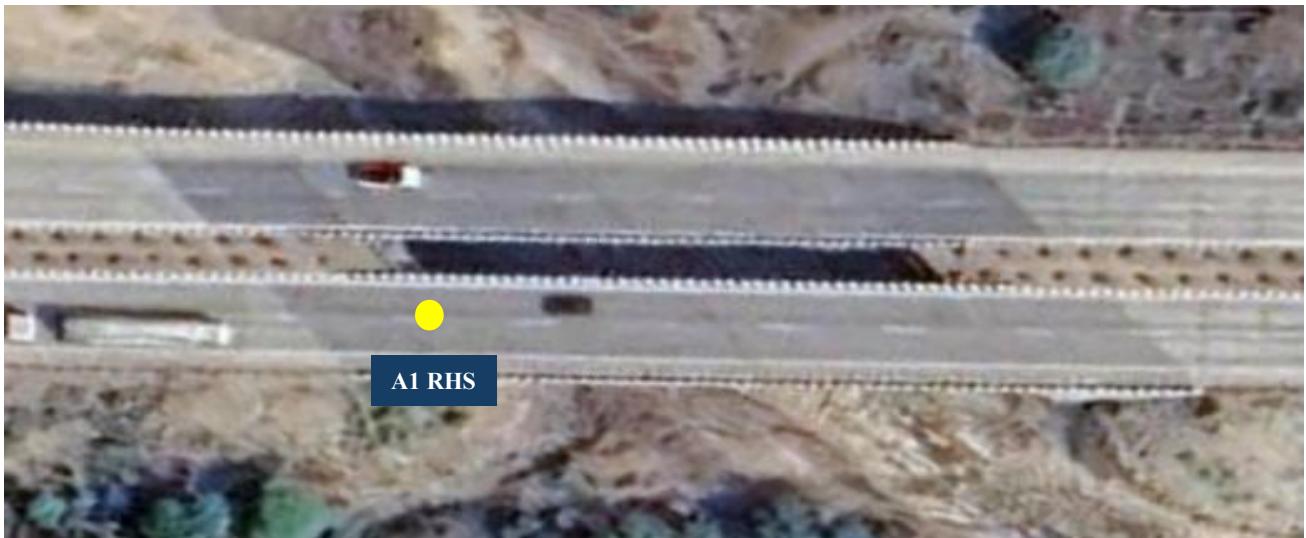
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** fluctuates around the zero line but is predominantly negative, ranging from approximately +4 mm to -14 mm.
- **Horizontal displacement (VH)** shows a clear and progressive upward trend, increasing from around +3–5 mm in late January to 22–23 mm by late June.

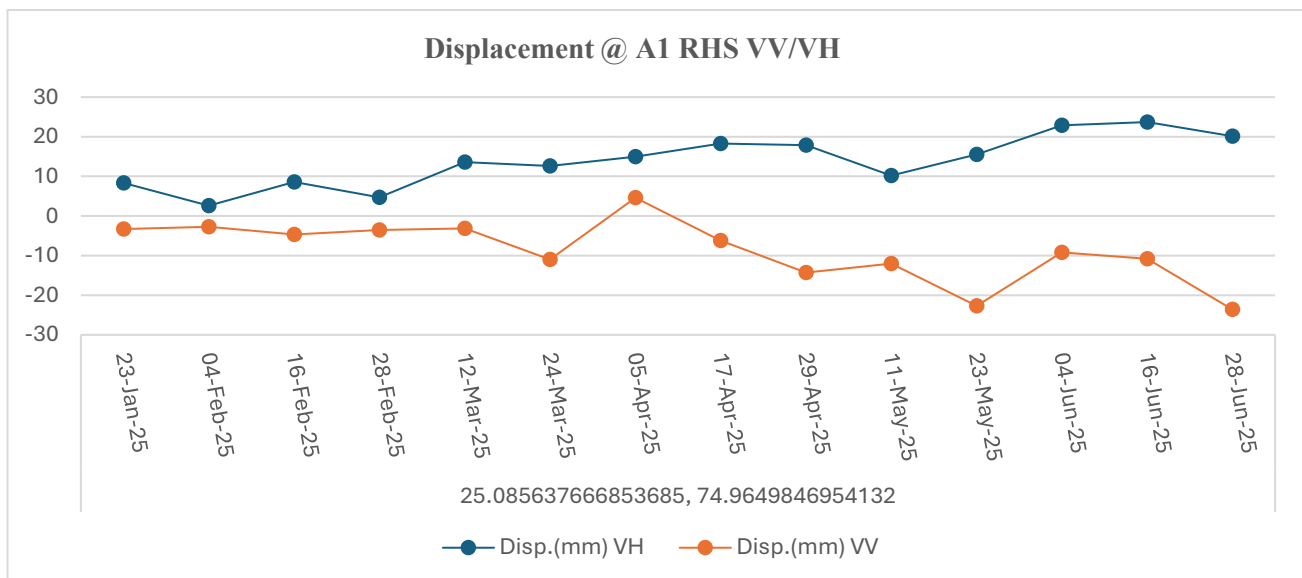
Critical Events & Anomalies:

- **February 2025:** VH drops briefly to 0 mm, while VV rises to +4 mm, indicating a short-term decoupled response with minimal lateral movement and slight vertical uplift.
- **May 2025:** VH remains elevated (13–15 mm), indicating sustained lateral displacement activity. VV drops to around -4 mm, reflecting renewed downward vertical movement during this period.
- **June 2025:** VH reaches its maximum recorded value (22–23 mm), while VV rebounds sharply to +6 mm.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.085637666853685, 74.9649846954132)



Key Observations:

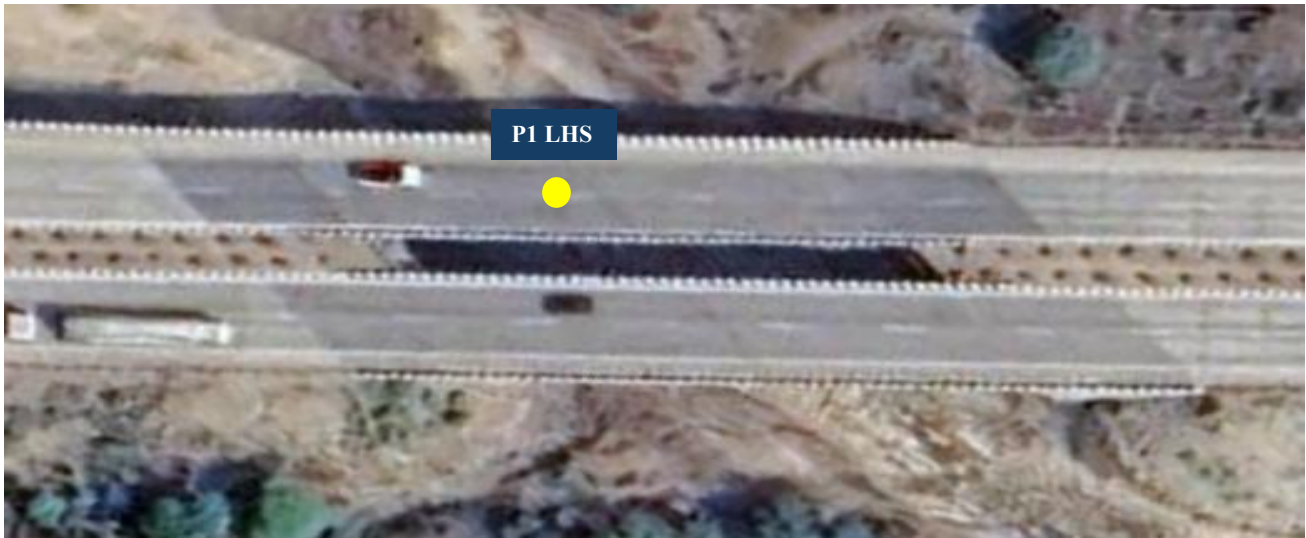
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows a strong and sustained upward trend, increasing from about +7–9 mm in late January to 22–24 mm by mid-June, before slightly reducing toward 20 mm at the end.
- **Vertical displacement (VV)** remains consistently negative throughout the monitoring period, ranging approximately from –3 mm to –25 mm, indicating continuous downward settlement at A1 RHS.

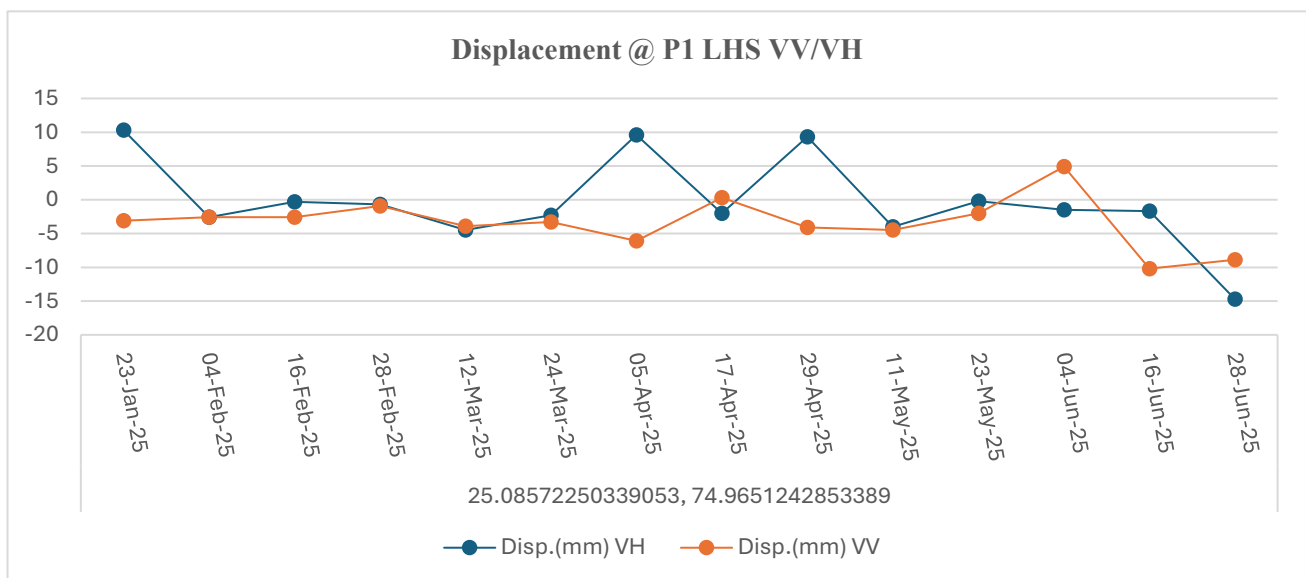
Critical Events & Anomalies:

- **February 2025:** VH fluctuates between 3 mm and 9 mm, while VV stays mildly negative (–3 to –5 mm), indicating early-stage minor deformation with limited structural response.
- **June 2025:** VV drops again to –25 mm, matching the maximum settlement magnitude, while VH slightly reduces to 20 mm, indicating settlement dominance toward the end of the monitoring cycle.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.08572250339053, 74.9651242853389)



Key Observations:

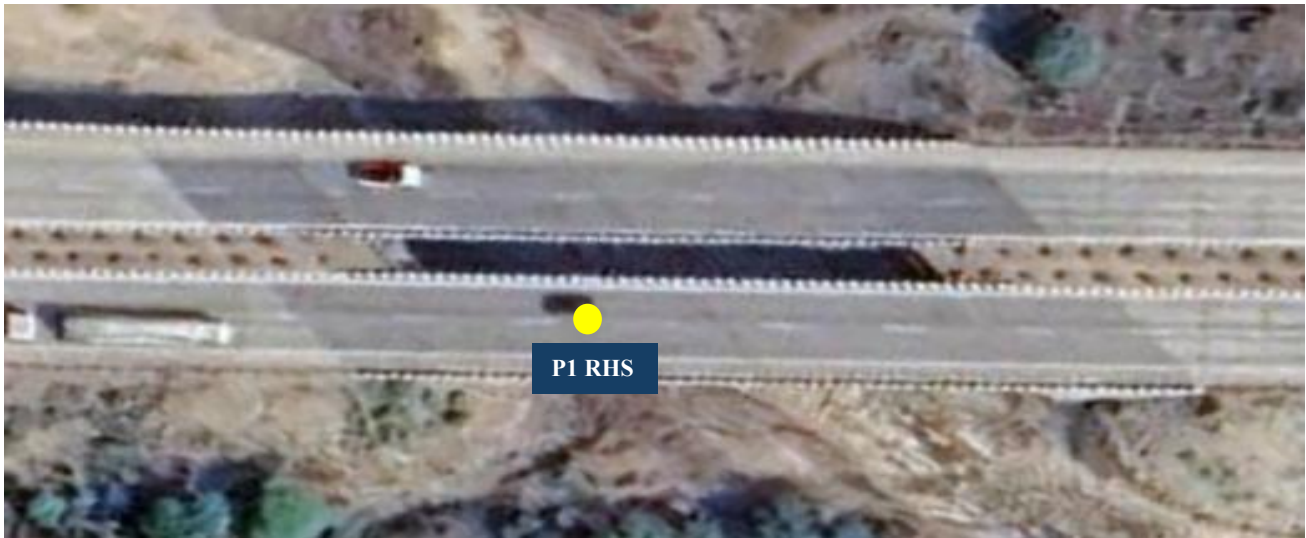
Displacement Magnitude Patterns:

- **Horizontal displacement** shows alternating positive and negative behaviour, ranging from about +10 mm to –15 mm.
- **Vertical displacement (VV)** fluctuates around the zero line but is predominantly negative, ranging approximately from –2 mm to –10 mm for most of the monitoring period.

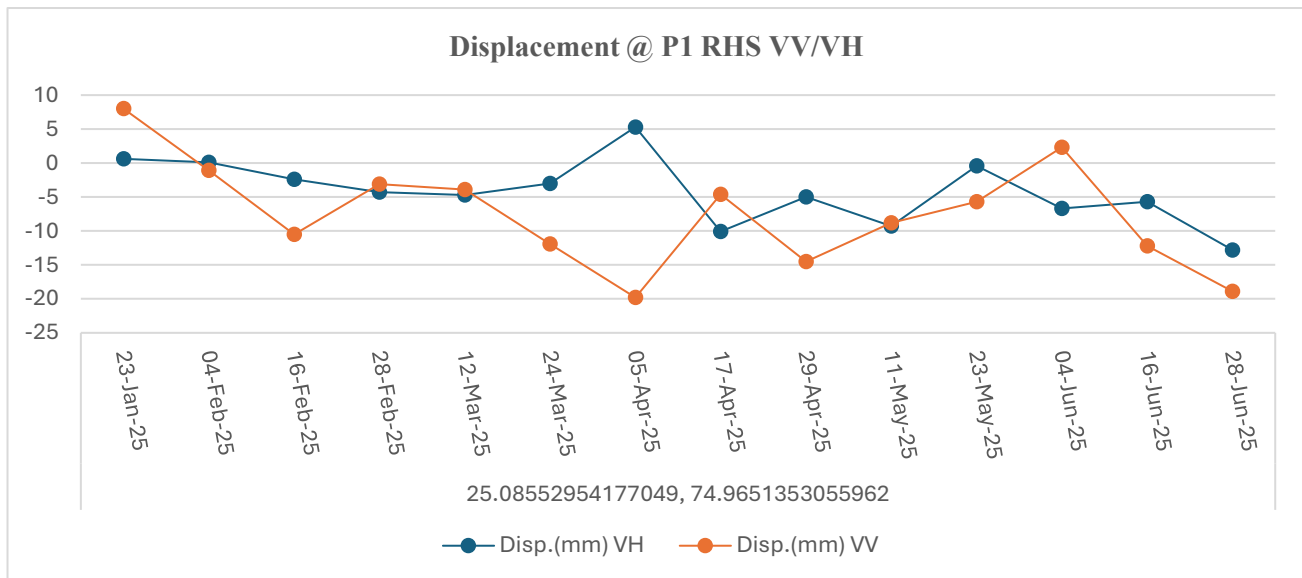
Critical Events & Anomalies:

- **January 2025:** VH records an early high peak (+10 mm), while VV is moderately negative (–4 mm). This represents an initial lateral displacement event without corresponding vertical uplift.
- **June 2025:** VH drops sharply to its lowest value (–15 mm on 28-Jun), representing the maximum lateral reversal event. VV concurrently falls to –9 to –10 mm, indicating a combined settlement and lateral shift, closing the monitoring period with intensified deformation.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.08552954177049, 74.9651353055962)



Key Observations:

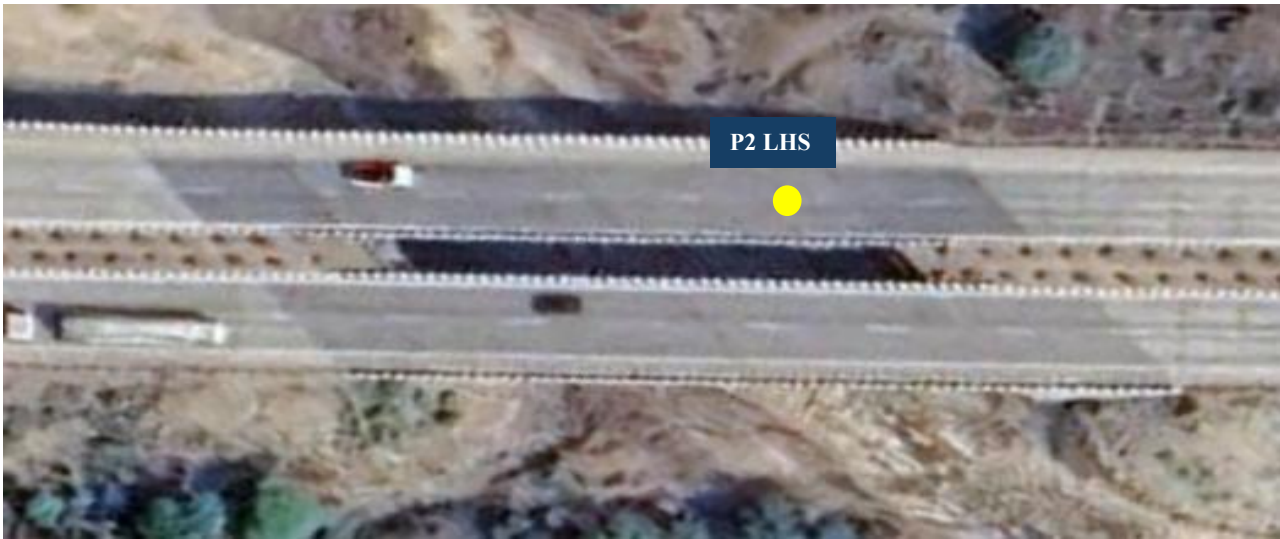
Displacement Magnitude Patterns:

- **Horizontal displacement** fluctuates within a moderate range, generally between +5 mm and –13 mm. Unlike abutment locations, VH does not display a monotonic trend, instead reflecting localized and reversible lateral movements typical of pier response.
- **Vertical displacement (VV)** shows large variability and predominantly negative values, ranging from approximately +8 mm to –20 mm.

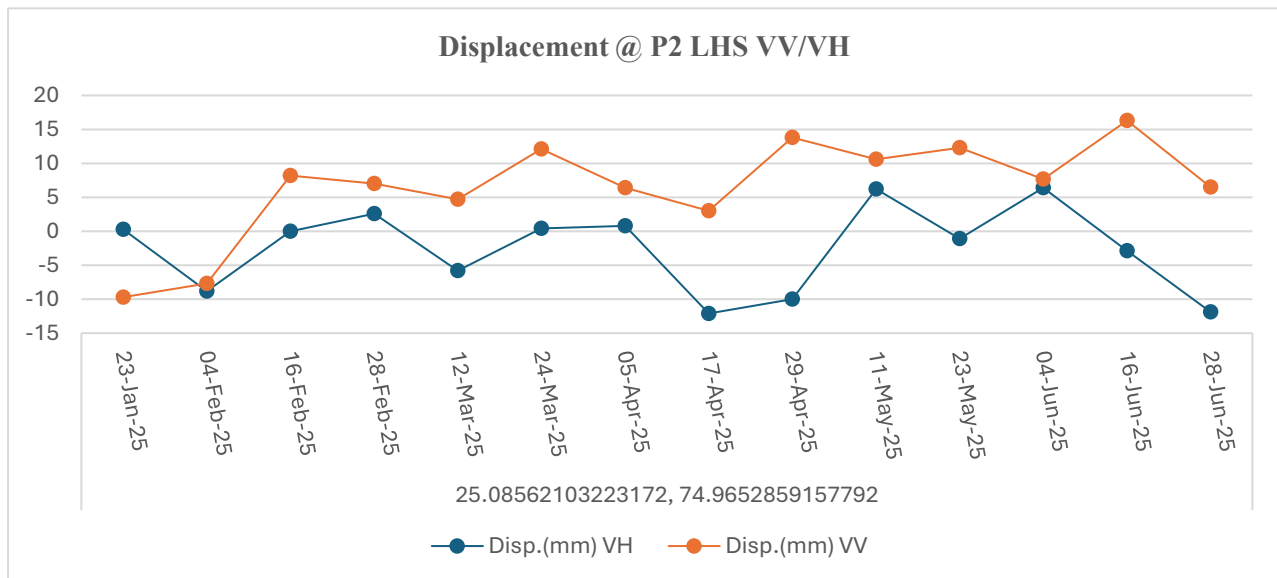
Critical Events & Anomalies:

- **August 2024:** VV records a brief negative dip to around –5 mm, representing the first notable vertical displacement anomaly despite otherwise positive upward movement.
- **March 2025:** VH decreases to around 8–10 mm, while VV remains elevated at about 14 mm, suggesting decoupled vertical and horizontal movement during this interval.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.08562103223172, 74.9652859157792)



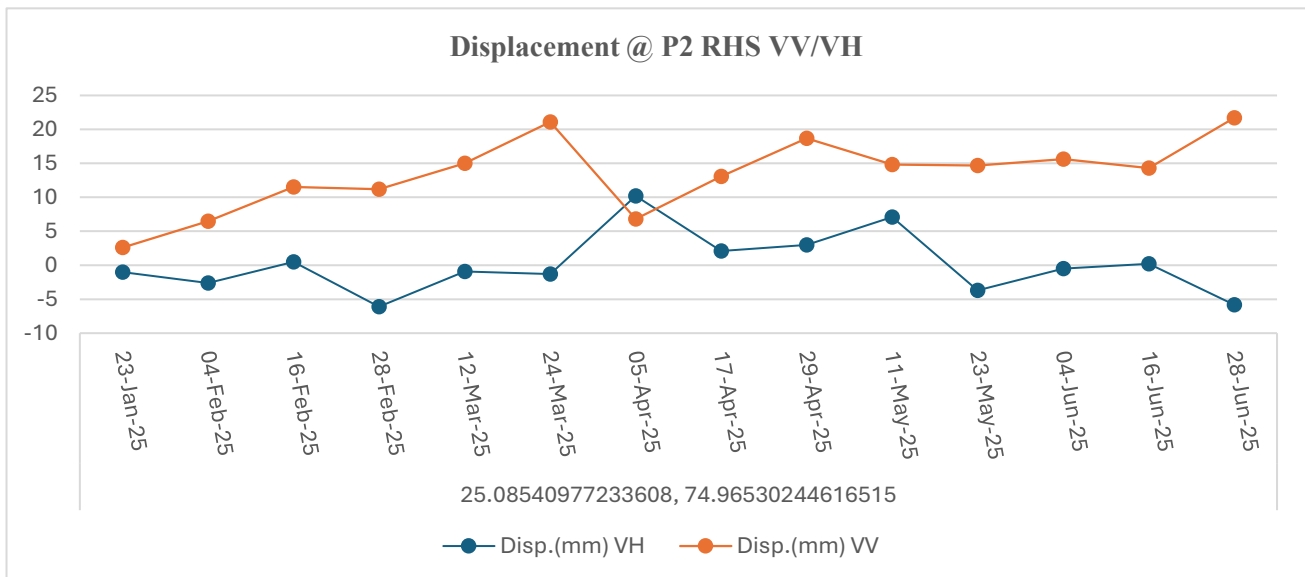
Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement** The absence of a consistent trend suggests localized lateral response, likely influenced by transient loading or temperature effects rather than progressive drift.
- **Vertical displacement (VV)** remains predominantly positive, ranging approximately from +2 mm to +16 mm, indicating overall upward vertical movement at P2 LHS. Although short-term fluctuations are present, the general behaviour reflects repeated uplift cycles rather than settlement.

Critical Events & Anomalies:

- **February 2025:** VV starts negative (−10 mm) but recovers rapidly to −7 mm, while VH shifts from 0 mm to −8 mm, marking an early adjustment phase with opposing vertical and horizontal movements.
- **June 2025:** VH drops again to −12 mm, closing the period with a significant lateral reversal. VV remains positive (+6–7 mm), confirming that vertical uplift persists despite lateral fluctuations.



Key Observations:

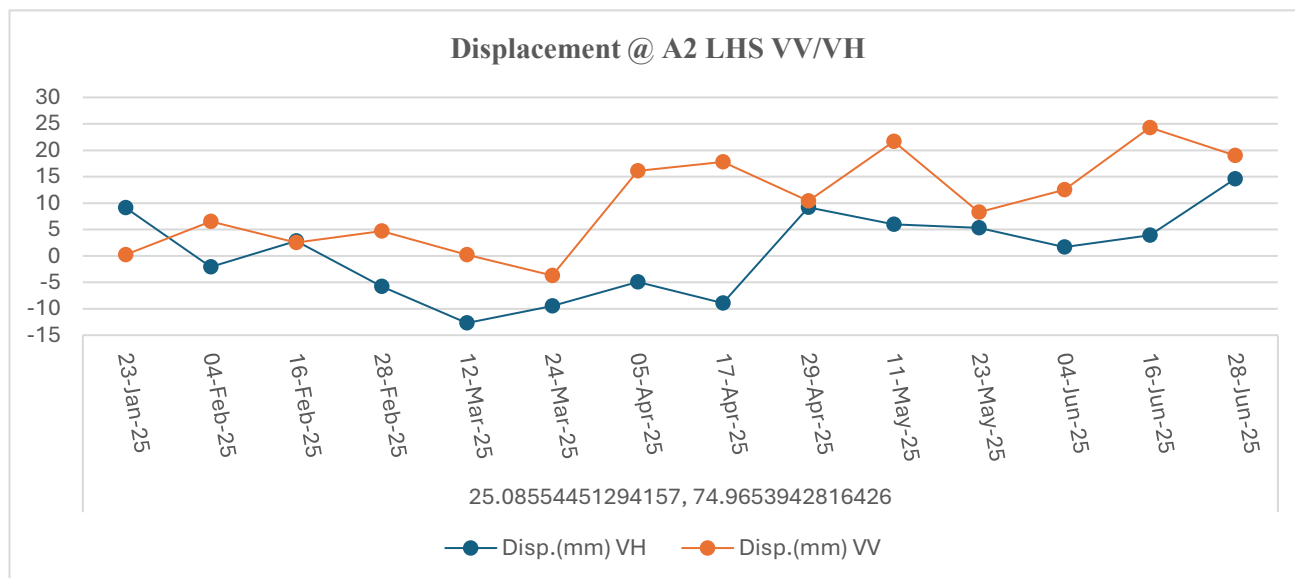
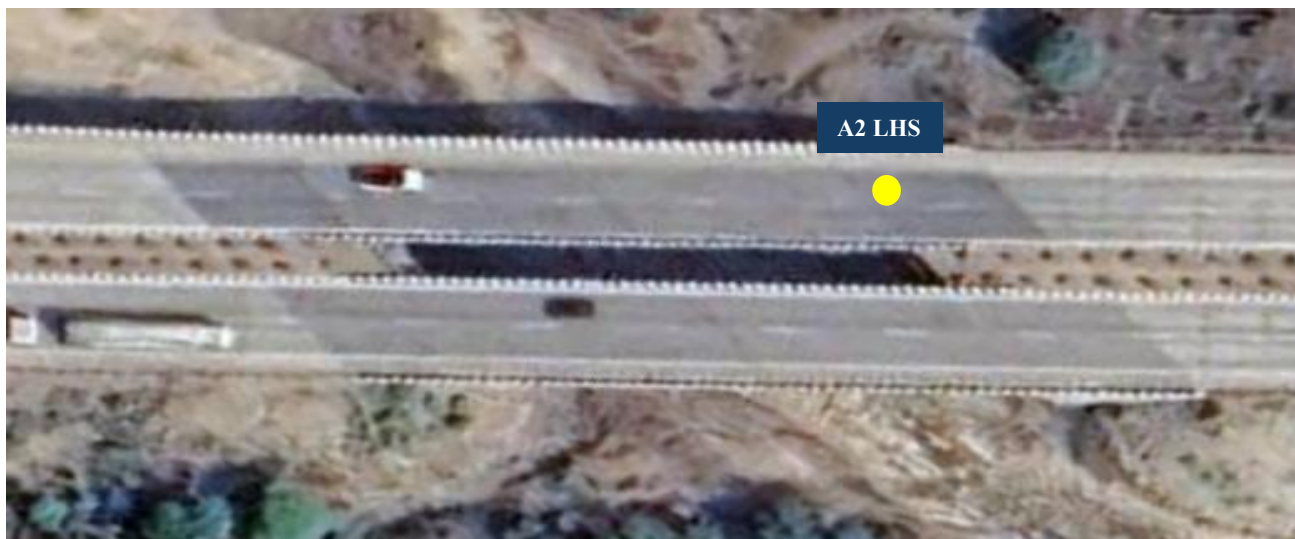
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** exhibits moderate variability with alternating direction, fluctuating between approximately +10 mm and –6 mm. VH does not show a long-term drift, instead reflecting localized and reversible lateral movements typical of pier behaviour.
- **Vertical displacement (VV)** remains consistently positive throughout the monitoring period, ranging approximately from +3 mm to +22 mm, indicating sustained upward vertical movement at P2 RHS.

Critical Events & Anomalies:

- **Jan-Feb 2025:** V starts at a low positive value (+3 mm) and begins a steady rise. VH remains close to zero to mildly negative (0 to –2 mm), indicating initial vertical response with minimal lateral activity.
- **June 2025:** VV reaches its maximum value (+22 mm), closing the monitoring period with strong upward vertical displacement

3.9 SAR Based Analysis at Abutment 2 LHS:



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH):** exhibits large variability with both positive and negative values, ranging from about +9 mm to –13 mm.
- **Vertical displacement (VV):** remains consistently positive with increasing magnitude, ranging approximately from 0 mm to +24 mm.

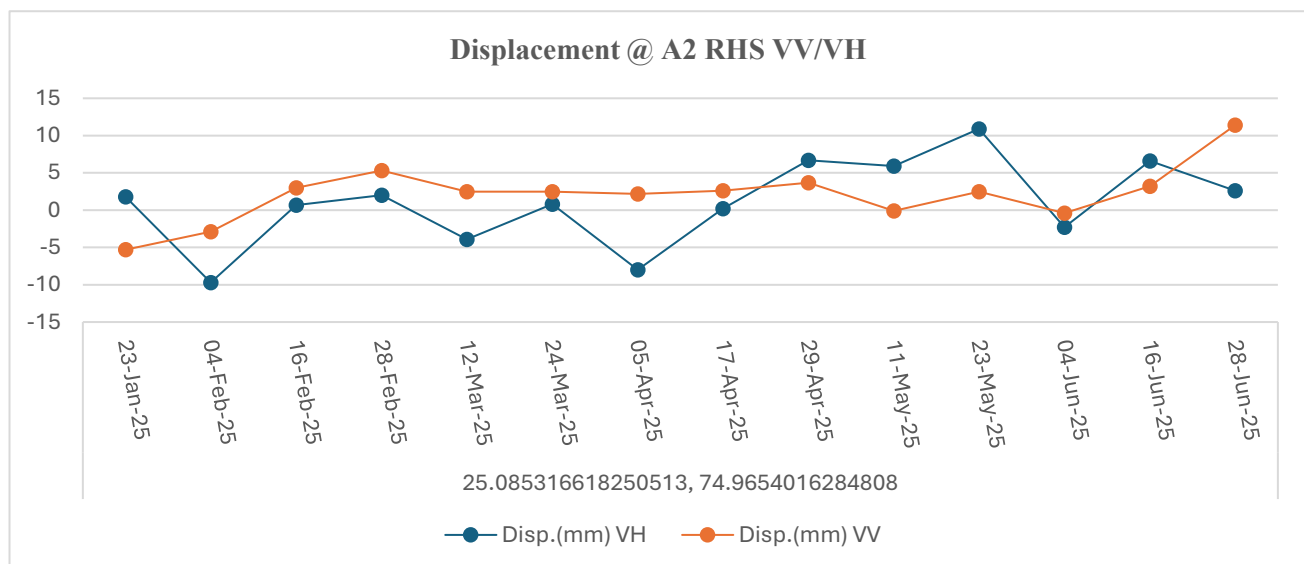
Critical Events & Anomalies:

- **January-February 2025:** VH starts positive (+9 mm) but quickly drops to –2 mm, indicating an early lateral reversal.
- **June 2025:** VH reaches its maximum positive value (+14–15 mm), closing the monitoring period with strong lateral displacement. VV remains elevated (+18–19 mm), confirming persistent uplift dominance.

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.085316618250513, 74.9654016284808)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH):** exhibits frequent directional changes, fluctuating between approximately +11 mm and -10 mm. Unlike a monotonic trend, VH displays cyclic lateral behaviour, suggesting localized response rather than cumulative lateral drift.
- **Vertical displacement (VV):** shows moderate variability with a generally positive tendency, ranging approximately from -5 mm to +11 mm.

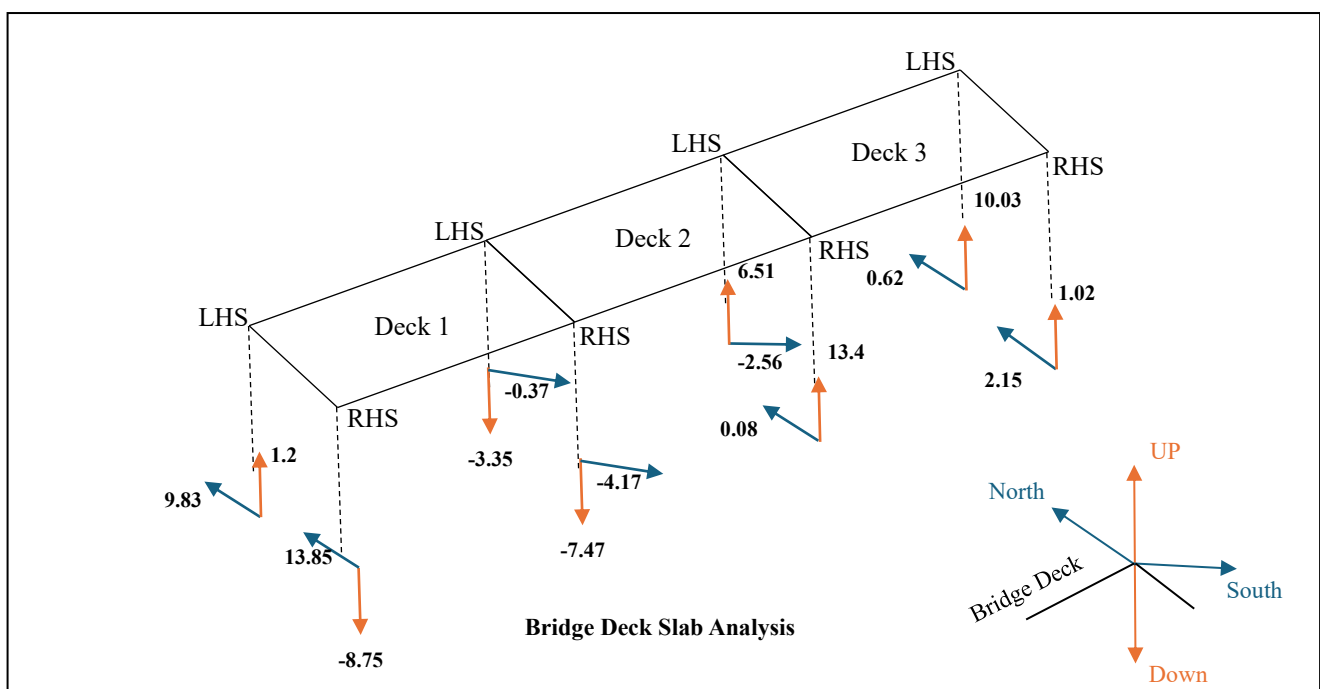
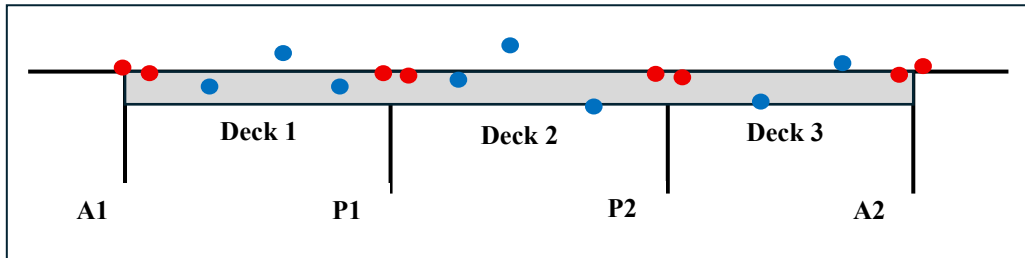
Critical Events & Anomalies:

- **February 2025:** VH drops sharply to -10 mm, marking the largest negative horizontal displacement in the dataset. VV remains mildly negative (-3 mm), indicating a predominantly lateral adjustment event.
- **June 2025 :** VV reaches its maximum value (+11 mm), closing the monitoring period with enhanced vertical uplift. VH settles near +2-3 mm, reflecting lateral stabilization.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The vertical displacement values across Decks 1, 2, and 3 show a combination of moderate downward settlements and localized upward movements. exhibits predominantly downward vertical displacements, with values reaching up to approximately -8.75 mm, indicating normal gravity-induced response under dead load and traffic effects.

Horizontal Displacement Behaviour:

- Horizontal displacements remain within acceptable engineering limits across all decks. Deck 1 shows moderate horizontal movements, with values up to approximately 13.85 mm, indicating normal longitudinal response to thermal expansion and contraction. Deck 2 displays relatively smaller horizontal shifts, mostly within 4 mm, reflecting stable deck behaviour and controlled lateral movement. Deck 3 exhibits minor northward horizontal trends, generally below 3 mm, which are gradual and uniform, consistent with natural thermal breathing of the bridge superstructure.

CONCLUSION:

- The combined vertical and horizontal displacement patterns across Decks 1, 2, and 3 indicate that the bridge deck slab system is performing as expected under operational loading, thermal variations, and seasonal effects. No abnormal displacement trends or signs of structural distress are evident from the observed values.